

ODD-OUTDEGREE TRANSPOSE DYNAMICS ON LOOPED DIGRAPHS: PARITY COLLAPSE, FIBRES, AND COMPONENTS

ANONYMOUS

ABSTRACT. On a labelled looped digraph, reverse every arc and then toggle every ordered pair, loops included, inside the set of vertices of odd current outdegree. Equivalently, on $M_n(\mathbb{F}_2)$ we iterate $\Phi(A) = A^\top + (A\mathbf{1})(A\mathbf{1})^\top$. We prove that one step maps onto the even-total-parity hyperplane, which is the complete recurrent set, and that all recurrent periods divide four. Every codomain fibre has size 0, 1, or $2^{n-1} + 1$, according to an explicit column-parity test. We then give exact formulas for fixed points, two- and four-cycles, depth-one states, and the finite dynamical zeta function. A second factorisation views the state-dependent left factor as an involutory transvection or a singular projection. Static graph complementation, binary-margin counting, and finite-map zeta bookkeeping are treated as background. The owner search is bounded; no novelty, priority, or external-release claim is made.

1. THE UPDATE AND ITS SUBTRACTION BOUNDARY

Fix $n \geq 1$, work over $\mathbb{F}_2$, and let $\mathbf{1}$ be the all-one column. A matrix $A \in M_n(\mathbb{F}_2)$ is the adjacency matrix of a labelled directed graph in which loops are allowed. Put

$$r = A\mathbf{1}, \quad c = A^\top\mathbf{1}, \quad \tau = \mathbf{1}^\top A\mathbf{1}.$$

Thus r and c are the out- and indegree parity vectors and $\tau = \mathbf{1}^\top r = \mathbf{1}^\top c$ is total arc parity. Define

$$(1) \quad \Phi(A) = A^\top + rr^\top.$$

The second summand toggles the complete looped subdigraph on the current odd-outdegree vertices. It is a rank-*at-most-one* correction and is recomputed after every step.

Local complementation and its binary-matrix formulation are classical [2, 6]; pivot and loop complementation have an established algebraic calculus [3]. Static subgraph complementation, including its rank and algorithmic interfaces, is also a separate mature operation [4, 5]. All such static operations, transpose, outer products, transvections, binary margin enumeration, and generic functional-graph bookkeeping receive zero contribution credit here. The residual question is only the exact temporal structure of the state-dependent update (1).

The internal subtraction is equally strict. Earlier portfolio systems use an involution norm, double adjugation, and a quadratic $\mathbb{F}_2$ -state shear. In particular, that shear admits period three and tail two, while (1) will have tail at most one and recurrent periods only 1, 2, 4. No generic matrix, quadratic-map, cycle, fibre, or zeta template is claimed. A bounded formula-and-owner search found no literal-map match; a bounded non-hit is not a novelty certificate.

2. THE PARITY QUOTIENT AND ALL ORBITS

For $B = \Phi(A)$, write its margins as $r(B), c(B), \tau(B)$.

Lemma 2.1 (quotient law). *For every A ,*

$$(2) \quad r(B) = c + \tau r, \quad c(B) = (1 + \tau)r, \quad \text{quad}\tau(B) = 0.$$

Consequently the even-total hyperplane $\mathcal{H}_n = \{A : \tau(A) = 0\}$ is invariant; on it the margin pair is swapped.

Proof. Multiplying (1) on the right by $\mathbf{1}$ gives $r(B) = c + r(r^\top\mathbf{1}) = c + \tau r$. Transposing first gives $c(B) = r + r(r^\top\mathbf{1}) = (1 + \tau)r$. Finally, $\tau(B) = \mathbf{1}^\top(c + \tau r) = \tau + \tau^2 = 0$ in $\mathbb{F}_2$. $\square$

Theorem 2.2 (complete temporal classification). *The image and complete recurrent set are both $\mathcal{H}_n$. Every state outside $\mathcal{H}_n$ has exact depth one. On $\mathcal{H}_n$,*

$$(3) \quad \Phi^2(A) = A + rr^\top + cc^\top, \quad \Phi^4(A) = A.$$

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If $r \neq c$, then A has exact period four. If $r = c$, its period is one or two.

Proof. The last identity in (2) shows that $\text{im } \Phi \subseteq \mathcal{H}_n$. The reverse inclusion follows from the explicit preimage in theorem 3.1. On $\mathcal{H}_n$ the margins swap, so a second use of (1) yields the first identity in (3); this added matrix has zero row and column margins. Repeating the same two steps cancels it in characteristic two, proving the second identity. Thus every point of $\mathcal{H}_n$ is recurrent. Every odd-total point reaches it after one step and cannot already be recurrent, proving exact depth one.

If $r \neq c$, then $rr^\top \neq cc^\top$ because the diagonal of $vv^\top$ is v ; hence $\Phi^2(A) \neq A$, and the period is exactly four. If $r = c$, (3) gives $\Phi^2(A) = A$. $\square$

Here depth means the least entrance time into the recurrent set. On a finite map this equals the distance to the first periodic point. The unique matrix in $M_0(\mathbb{F}_2)$ may separately be declared fixed; all remaining formulas assume $n \geq 1$.

3. EVERY ONE-STEP FIBRE AND A SECOND ROUTE

Theorem 3.1 (codomain-wide fibre trichotomy). *For every $B \in M_n(\mathbb{F}_2)$,*

$$(4) \quad |\Phi^{-1}(B)| = \begin{cases} 0, & \tau(B) = 1, \\ 1, & \tau(B) = 0 \text{ and } c(B) \neq 0, \\ 2^{n-1} + 1, & \tau(B) = 0 \text{ and } c(B) = 0. \end{cases}$$

In particular, $|\text{im } \Phi| = 2^{n^2-1}$, and precisely $2^{n(n-1)}$ image targets have the large fibre.

Proof. If a preimage has row margin r , solving (1) for it gives $A = B^\top + rr^\top$. Requiring $A\mathbf{1} = r$ is equivalent to

$$(5) \quad r = c(B) + (\mathbf{1}^\top r)r.$$

For even-weight r , this forces $r = c(B)$ and exists exactly when $\tau(B) = 0$; it then supplies one preimage. For odd-weight r , (5) is equivalent to $c(B) = 0$. In that case all 2^{n-1} odd vectors work, in addition to the even solution $r = 0$. This proves (4) and also $\text{im } \Phi = \mathcal{H}_n$.

There are 2^{n-1} choices for each even-parity column, independently over n columns, so exactly $2^{n(n-1)}$ matrices have $c(B) = 0$. $\square$

The odd-total zero case in (4) is essential. For example, when $n = 1$, the target $[1]$ has nonzero column margin but no preimage.

Proposition 3.2 (transvection–projection factorisation). *The same collapse has the factorisation*

$$(6) \quad \Phi(A) = (I + r\mathbf{1}^\top)A^\top.$$

If $\tau = 0$, the left factor is an involution. If $\tau = 1$, it is an idempotent rank- $n - 1$ projection with kernel $\langle r \rangle$ and image $\mathbf{1}^\perp$.

Proof. Since $\mathbf{1}^\top A^\top = r^\top$, (6) is (1). Put $L_r = I + r\mathbf{1}^\top$. Direct multiplication gives $L_r^2 = I + \tau r\mathbf{1}^\top$. This is I when $\tau = 0$ and L_r when $\tau = 1$. In the latter case $L_r r = 0$ and $\mathbf{1}^\top L_r x = 0$ for every x ; the rank is therefore $n - 1$. $\square$

This factorisation independently explains why the odd coset is singularly projected to one parity hyperplane, whereas the direct margin calculation determines individual fibres.

4. COMPONENTS AND ZETA FUNCTION

Lemma 4.1 (prescribed margins). *For $u, v \in \mathbb{F}_2^n$, there is a matrix with row margin u and column margin v exactly when $\mathbf{1}^\top u = \mathbf{1}^\top v$. In that case there are $2^{(n-1)^2}$ such matrices.*

Proof. Choose the upper-left $(n-1) \times (n-1)$ entries freely. The prescribed row sums force the last entry of each of the first $n-1$ rows, and the prescribed column sums force the last entry of each of the first $n-1$ columns. The two requirements on the bottom-right corner agree precisely when the total parities agree. $\square$

Put

$$F_n = 2^{n(n-1)/2}, \quad C_{2,n} = \frac{2^{n(n-1)} - F_n}{2}, \quad C_{4,n} = \frac{2^{n^2-1} - 2^{n(n-1)}}{4}.$$

Theorem 4.2 (functional components). *The map has F_n fixed points, $C_{2,n}$ two-cycles, and $C_{4,n}$ four-cycles. It has 2^{n^2-1} depth-one states. With the usual finite-map convention [1],*

$$(7) \quad \zeta_{\Phi_n}(z) = (1-z)^{-F_n}(1-z^2)^{-C_{2,n}}(1-z^4)^{-C_{4,n}}.$$

Proof. A fixed point satisfies $A + A^T = rr^T$. Comparing diagonals forces $r = 0$; then A is symmetric with zero row sums. Its off-diagonal entries are arbitrary and its diagonal is forced by the row sums, giving dimension $n(n-1)/2$ and hence F_n fixed points.

By lemma 4.1, for each of the 2^{n-1} even vectors r , there are $2^{(n-1)^2}$ recurrent matrices with margins (r, r) . Thus $2^{n(n-1)}$ recurrent points have period dividing two. Removing fixed points and dividing by two gives $C_{2,n}$. All remaining recurrent points have exact period four by theorem 2.2, giving $C_{4,n}$. The odd coset has size 2^{n^2-1} and consists exactly of the depth-one points. Finally, the standard product of one factor $(1-z^k)^{-1}$ per k -cycle gives (7). $\square$

5. EXACT CONTROL AND LIMITATIONS

The accompanying deterministic verifier exhausts every one of the 2^{n^2} matrices for $1 \leq n \leq 4$. It checks both forms of the update, the quotient and fourth-iterate identities, exact depths and periods, all codomain fibres including zero fibres, the affine margin count, and every component formula. These bounded computations are counterexample searches, not proofs of the all- n statements above.

The result is specific to the binary field, square matrices, transpose, and the recomputed row-margin selector. Rectangular carriers, other fields, other bilinear forms, and static graph complementation require new proofs and owner gates. Literature screening was bounded and can establish a hit but not an absence. Accordingly we make no claim of novelty, priority, authorship, posting readiness, or external release.

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